Unit no. ° 14

PROBABILITY: CONDITIONING AND WEIGHTED TREES

26 April to 24 May 2027

I Sample space, outcomes and events

Vocabulary

A random experiment has several possible outcomes, and we do not know in advance which one will be obtained. An outcome is a possible result; the sample space is the set of all outcomes. An event is a subset of the sample space; an elementary event contains a single outcome. The impossible event is $\varnothing$; the sure event is Omega.

Operations on events

$\overline{A}$ is the complementary event of A. $A \cap B$ means "A and B"; $A \cup B$, "A or B", possibly both. Two events are mutually exclusive if $A \cap B = \varnothing$; otherwise, they are not disjoint.

Example 14.1

A six-sided die is rolled. Write ?, A : "obtain an even number", B : "obtain 4", then $\overline{A}$, $A \cap B$ and $A \cup B$. A, B : are they compatible?

Solution:

$\Omega = \{1;2;3;4;5;6\}$, $A = \{2;4;6\}$, $B = \{4\}$, $\overline{A} = \{1;3;5\}$, $A \cap B = \{4\}$, $A \cup B = A$. The events are compatible because they share outcome 4.

Example 14.2

With $A = \{a;b;c;d;e\}$ and $B = \{b;e;f;g;h\}$, give the intersection and the union. Can we determine $\overline{A}$ without specifying the sample space?

Solution:

$A \cap B = \{b;e\}$ and $A \cup B = \{a;b;c;d;e;f;g;h\}$. The complement depends on the sample space; we cannot determine it without this information.

II Probability model and equally likely outcomes

Probability in a finite universe

A model assigns to each outcome a probability that is positive or zero; the sum is 1. The probability of an event is the sum of the probabilities of its outcomes. We have

$$P(\Omega) = 1, \quad P(\varnothing) = 0, \quad P(\overline{A}) = 1 - P(A).$$

With equally likely outcomes, all outcomes have the same probability and

$$P(A) = \frac{\text{Card}(A)}{\text{Card}(\Omega)}.$$

Equally likely outcomes are an assumption of the model, not a consequence of the list of outcomes alone.

Example 14.3

Assume the die is fair. For A = "even" and C = "greater than or equal to 5", calculate $P(A)$, $P(C)$, $P(A \cap C)$ and $P(A \cup C)$.

Solution:

$P(A) = 3/6 = 1/2$, $P(C) = 2/6 = 1/3$, $A \cap C = \{6\}$ so $P(A \cap C) = 1/6$. $A \cup C = \{2; 4; 5; 6\}$, therefore $P(A \cup C) = 4/6 = 2/3$. Outcome 6 must not be counted twice.

Example 14.4

A drawing pin can land point up or point down. Can we deduce that each outcome has probability 1 /2 ? How can we propose a model?

Solution:

No: having two outcomes does not make them equally likely. We can repeat the tosses in comparable conditions and use the observed frequencies to propose a model, bearing in mind the fluctuations and the characteristics of the object.

Guidance for the teacher

Revisit the link between observed frequency and assumed probability. An experimental result does not prove that the outcomes are exactly equally likely.

III Conditional probability from a table

Definition

For $P(A) > 0$, the probability of B given A is

$$P_A(B) = \frac{P(A \cap B)}{P(A)}, \qquad P(A \cap B) = P(A)P_A(B).$$

In an equally likely draw from a finite population:

$$P_A(B) = \frac{\text{Card}(A \cap B)}{\text{Card}(A)}.$$

We also write $P(B \mid A)$. $P_A(B)$ and $P_B(A)$ are generally not equal.

Example 14.5

A pupil is drawn uniformly at random from the class of 32 pupils studied earlier. G : "boy", N : "takes NSI".

	G	$\overline{G}$	Total
N	12	9	21
$\overline{N}$	6	5	11
Total	18	14	32

Calculate $P(G)$, $P(N)$, $P(G \cap N)$, $P_N(G)$ and $P_G(N)$.

Solution:

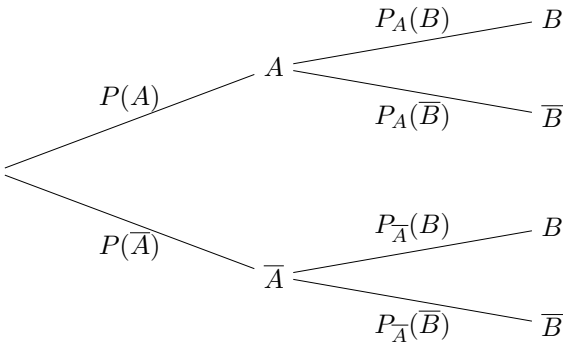
$P(G) = 18/32 = 9/16$, $P(N) = 21/32$, $P(G \cap N) = 12/32 = 3/8$. $P_N(G) = 12/21 = 4/7$, while $P_G(N) = 12/18 = 2/3$. By the formula: $P_N(G) = (12/32)/(21/32) = 12/21$. Given that the pupil takes NSI, the reference population is reduced to the 21 pupils concerned.

Watch out. Conditioning requires the known event to have non-zero probability; we do not divide by zero.

IV Weighted trees and the probability of a path

Rules

At each node, the sum of the probabilities on the outgoing branches is 1. A branch of the second level carries a conditional probability. A path represents an intersection of events; its probability is the product of the probabilities written on this path.



Example 14.6

We know $P(A) = 0{,}6$, $P_A(B) = 0{,}7$ and $P_{\overline{A}}(\overline{B}) = 0{,}9$. Complete the other branches and calculate the probabilities of the four paths.

Solution:

$P(\overline{A}) = 0{,}4$, $P_A(\overline{B}) = 0{,}3$, $P_{\overline{A}}(B) = 0{,}1$.

$$P(A \cap B) = 0{,}6 \times 0{,}7 = 0{,}42, \quad P(A \cap \overline{B}) = 0{,}18,$$

$$P(\overline{A} \cap B) = 0{,}04, \qquad P(\overline{A} \cap \overline{B}) = 0{,}36.$$

The sum of the four probabilities is 1, which checks the consistency of the model.

Example 14.7

Justify multiplying along the path A then B from the definition of $P_A(B)$.

Solution:

Since $P_A(B) = P(A \cap B)/P(A)$ and $P(A) > 0$, multiplying by $P(A)$ gives $P(A)P_A(B) = P(A \cap B)$.

V Moving from a table to a tree

Example 14.8

For the class of 32 pupils, draw a tree starting with NSI / no NSI, then boy / girl.
Then construct the tree in the reverse order. Check, in both trees, the probability of choosing a girl given NSI.

Solution:

First tree: initial branches $P(N) = 21/32$, $P(\overline{N}) = 11/32$. After N : $P_N(G) = 12/21 = 4/7$, $P_N(\overline{G}) = 9/21 = 3/7$. After $\overline{N}$: $6/11$ and $5/11$. Second tree: initial branches $P(G) = 18/32 = 9/16$, $P(\overline{G}) = 14/32 = 7/16$. After G : $P_G(N) = 12/18 = 2/3$, then $1/3$. After $\overline{G}$: $P_{\overline{G}}(N) = 9/14$, then $5/14$. The required path has probability $(21/32)(9/21) = 9/32$ in the first tree and $(14/32)(9/14) = 9/32$ in the second. Changing the order preserves the probabilities of the intersections but changes the probabilities of the branches.

Example 14.9

An urn contains 3 red balls and 2 blue ones. We draw two balls in succession **without replacement**. Construct the colour tree and calculate the probability of drawing two reds.

Solution:

First level: R with $3/5$, B with $2/5$. After R, 2 red and 2 blue remain: branches $1/2, 1/2$. After B, 3 red and 1 blue remain: branches $3/4, 1/4$. Hence $P(R_1 \cap R_2) = (3/5)(1/2) = 3/10$. Using $3/5$ on the second red branch would ignore that the draw is without replacement.

VI Reversing the conditioning and false positives

Example 14.10: Fictional component inspection

A batch contains 10000 components, 100 of which are defective. A test is positive for 99 of the defective components and for 495 of the non-defective components. A component is chosen uniformly at random. D : " defective"; T: "positive test".

1. Complete the two-way table: test / true status.

2. Calculate $P_D(T)$ and $P_T(D)$.

3. Give the number of false positives and false negatives.

4. Calculate sensitivity $P_D(T)$ and specificity $P_{\overline{D}}(\overline{T})$.

Solution:

	D	$\overline{D}$	Total
T	99	495	594
$\overline{T}$	1	9405	9406
Total	100	9900	10000

$P_D(T) = 99/100$ = 99%. However $P_T(D) = 99/594 = 1/6 \approx 16{,}67\%$. A positive test does not imply here, therefore, a 99% probability of a defect. The 495 false positives are the sound components that test positive; the false negative is the defective component that tests negative. The sensitivity is 99% ; specificity 9405 /9900 = 95%. The low initial proportion of defective components explains why there are more false positives than true positives despite the good sensitivity.

Read the test result

"Positive among the defective" and "defective among the positives" are two different conditionings. Always name the reference group before calculating.

Guidance for the teacher

The numbers are fictitious. The table makes it possible to study the reversal without requiring a general formula of total probability: this is not expected in the 2026 Seconde curriculum.

VII Simulation and law of large numbers

Simplified version

When an experiment is repeated a large number of times in the same model with independent repetitions, the observed frequency of an event is, except in rare cases, close to its probability.
It fluctuates: it does not necessarily get closer to it with each new experiment.

```
from random import Random

def frequences_six(n, graine=1):
    # n integer >= 1 ; model of independent tosses
    rng = Random(graine)
    succes = 0
    frequences = []
    for k in range(1, n+1):
        if rng.randint(1, 6) == 6:
            succes = succes+1
        frequences.append(succes/k)
    return frequences
```

Example 14.11

Plot the frequencies for $n = 1000$, with several seeds. Compare with $1/6$. Why fix the seed? Can we require the final frequency to be exactly $1/6$?

Solution:

The plots depend on the seed and fluctuate around $1/6$. Fixing the seed makes the simulation reproducible. With 1000 rolls, a final frequency of exactly $1/6$ is actually impossible, since $1000/6$ is not an integer. A frequency close to it is consistent with the model, but is not a proof.

Example 14.12

Modify the program to track the frequency of the event "even result". What reference probability to use?

Solution:

Replace the test with `rng.randint(1, 6) % 2 == 0`. The reference probability is $3/6 = 1/2$. Each experiment must make a single draw; do not call the generator several times for testing the same result.